Anatomy & Physiology — Body Systems Sheets

Structure–function tables, physiology essentials, clinical links and exam traps, system by system

MindShelf

How to Use These Sheets

Every body system follows the same four-part layout:

1. **Key structures → function:** a table that pairs each structure with what it does.
2. **Physiology essentials:** the mechanisms behind the table.
3. **Clinical link:** one condition that shows the physiology at work, with the source it comes from.
4. **Traps:** mix-ups that cost points on exams.

These sheets summarize core facts for review. They are not a full course and do not replace your textbook or your program's materials.

Levels of organization

Cells (the smallest units that can live and reproduce) → **tissues** (groups of similar cells and the material between them) → **organs** (several tissues working together, e.g., the stomach) → **organ systems** (organs that share a complex job).

Directions and planes

Term	Meaning
Superior (cranial) / inferior (caudal)	Toward the head / away from the head
Anterior (ventral) / posterior (dorsal)	Front / back of the body
Medial / lateral	Toward the midline / away from the midline
Proximal / distal	Closer to / farther from the trunk or a limb's point of origin
Sagittal plane	Vertical, front to back: splits right from left (median plane = exactly on the midline)
Coronal (frontal) plane	Vertical, side to side: splits anterior from posterior
Transverse (axial) plane	Horizontal: splits upper from lower parts

Integumentary System

Key structure	Function
Epidermis	Outer layer of keratinocytes; it has no blood vessels of its own and is nourished from the dermis
Stratum basale	Deepest epidermal layer: dividing basal cells, melanocytes (make melanin) and Merkel cells (touch)
Stratum spinosum	Keratinocytes plus Langerhans cells, which pick up antigens entering damaged skin and alert the immune system
Stratum granulosum, lucidum	Keratinocytes flatten and lose water as they are pushed toward the surface; the lucidum exists only in thick skin (palms, soles, digits)
Stratum corneum	Surface layer: 10 to 30 rows of dead keratinocytes forming the tough outer barrier
Dermis, papillary layer	Thin collagen fibers and small vessels that feed the epidermis and help regulate temperature
Dermis, reticular layer	Thicker, denser collagen: strength, structure and elasticity
Dermal structures	Hair follicles, sebaceous (oil) glands, sweat glands, nerve endings, blood vessels; arrector pili muscles raise goosebumps
Hypodermis (subcutis)	Network of fat and collagen crossed by blood vessels, lymph vessels and nerves

Physiology essentials

- **Functions:** barrier against pathogens, injury, heat and light; temperature control; sensation; storage of water and fat; vitamin D; a front line for the immune system.
- **Renewal:** new cells form in the stratum basale and are pushed upward, flattening and dying until they reach the stratum corneum.
- **Pigment:** melanocytes in the basal layer make melanin, which helps protect the DNA of epidermal cells from UV damage.
- **Heat:** evaporating sweat carries heat away; dermal arterioles dilate to shed heat and constrict in the cold to keep it.
- **Vitamin D:** with UV light the epidermis turns a cholesterol derivative into vitamin D_3; the liver and kidneys activate it, and it is needed to absorb calcium and phosphorus for bone.
- **Sensation:** Merkel cells and dermal receptors sense touch, temperature and pain; cells around hair follicles sense pressure.

CLINICAL NOTE Pressure injuries

Pressure injuries (bedsores) form when a person stays in one position too long, most often where bone lies close to the skin: heels, hips, ankles, elbows, back. People who are bedridden, use a wheelchair or cannot shift on their own are most at risk. Prevention: keep skin clean and dry, change position about every 2 h, use pressure-relieving cushions and surfaces. Source: MedlinePlus, *Pressure Sores* (updated Dec. 18, 2024).

TRAP

The epidermis has **no blood vessels**. A scrape that bleeds has reached the dermis.

TRAP

Melanocytes sit in the **stratum basale** of the epidermis, not in the dermis.

TRAP

Deep to surface: basale → spinosum → granulosum → lucidum → corneum. Thin skin has only 4 layers (no lucidum), thick skin 5. Questions often list them surface first.

Skeletal System

Key structure	Function
Compact bone	Dense bone built of osteons (haversian systems): a central canal with blood vessels ringed by lamellae of matrix
Lacunae and canaliculi	Small spaces that hold osteocytes, linked by tiny channels that carry nutrients
Spongy (cancellous) bone	Trabeculae (plates and bars) that follow lines of stress; the spaces between them hold red marrow
Red marrow	Hematopoiesis: forms blood cells
Osteoblasts	Form new bone
Osteocytes	Mature bone cells within the matrix
Osteoclasts	Resorb (break down) bone; their balance with osteoblasts maintains bone tissue
Axial skeleton	80 bones: skull (with the ear ossicles), hyoid, vertebral column, thoracic cage (sternum and ribs)
Appendicular skeleton	126 bones: pectoral girdle (clavicles, scapulae), upper limbs, pelvic girdle (hip bones), lower limbs
Synarthroses	Immovable joints, bones held by thin fibrous tissue (e.g., skull sutures)
Amphiarthroses	Slightly movable joints joined by cartilage (e.g., pubic symphysis, intervertebral disks, costal cartilages)
Diarthroses (synovial)	Freely movable, the most common adult joints: articular cartilage, joint cavity, fibrous capsule with ligaments, synovial membrane making lubricating fluid

Physiology essentials

- **Functions:** support against gravity; protection of the brain, spinal cord, heart and lungs; levers for movement with the muscles; calcium storage and release; blood cell formation in red marrow.
- **Count:** 80 axial + 126 appendicular = 206 bones in the adult skeleton.
- **Living tissue:** bone needs a blood supply, uses oxygen and nutrients, and remodels under mechanical stress; trabeculae realign when the direction of stress changes.
- **Calcium:** calcium is the most abundant mineral of the matrix. PTH raises blood Ca^{2+} and calcitonin lowers it (see Endocrine System).
- **Remodeling:** osteoblasts lay down bone and osteoclasts resorb it; the balance between them keeps bone mass stable.

CLINICAL NOTE Osteoporosis
Bones weaken and break more easily. It is usually silent until a fracture, most often of the hip, a vertebra or the wrist. Risk rises with female sex (especially after menopause), age over 50, a slim build, family history, long-term use of some medicines such as corticosteroids, smoking, heavy drinking and inactivity. Prevention relies on calcium and vitamin D in the diet and regular physical activity. Source: MedlinePlus, *Osteoporosis* (updated Sept. 30, 2025).

TRAP
Osteoblasts build, **osteoclasts** break down, **osteocytes** are the mature cells that stay in the matrix.

TRAP
The pectoral and pelvic **girdles** belong to the appendicular skeleton, not the axial one.

TRAP
Most adult joints are freely movable **synovial** joints; skull sutures are immovable (synarthroses).

Muscular System

Key structure	Function
Skeletal muscle	Striated, voluntary, many nuclei per fiber; attached to bones
Cardiac muscle	Striated, involuntary (autonomic control), one central nucleus per cell; wall of the heart
Smooth muscle	Not striated, involuntary, one central nucleus; walls of hollow organs and vessels (gut, bladder, uterus)
Connective sheaths	Epimysium wraps the whole muscle, perimysium each fascicle (bundle of fibers), endomysium each fiber
Tendon and aponeurosis	Extensions of the sheaths that attach a muscle to the periosteum of a bone or to another muscle
Neuromuscular junction	The motor neuron releases acetylcholine (ACh); the fiber depolarizes and the impulse runs into the T-tubules
Sarcoplasmic reticulum	Stores Ca^{2+}, releases it when the fiber is stimulated, then pumps it back (using ATP) so the fiber relaxes
Troponin and tropomyosin	Ca^{2+} binds troponin, which shifts tropomyosin off the binding sites on actin
Sarcomere (actin and myosin)	Myosin heads bind actin and pull the thin filaments toward the center: the sarcomere shortens, the filaments do not

Physiology essentials

- **Functions:** movement of bones, organs and vessels; posture; stability of the joints the tendons cross (knee, shoulder); heat, with nearly 85% of body heat coming from muscle contraction.
- **Excitation:** a fiber contracts only after a nerve impulse: ACh at the neuromuscular junction → action potential along the membrane and T-tubules → Ca^{2+} release.
- **Cross-bridge cycle:** Ca^{2+} on troponin uncovers actin → myosin heads attach and pull (power stroke) → ATP lets them let go and re-cock → the cycle repeats while Ca^{2+} and ATP last.
- **Relaxation:** ACh release stops, Ca^{2+} is pumped back into the sarcoplasmic reticulum, tropomyosin covers actin again.
- **Energy:** creatine phosphate covers about 15 s of work; glycolysis covers short intense bursts; aerobic respiration supplies about 95% of the ATP at rest or in moderate activity.

CLINICAL NOTE Myasthenia gravis
An autoimmune disease: antibodies disrupt signal transmission at the neuromuscular junction, so muscles do not get the nerve signal. Weakness of voluntary muscles worsens with activity and improves with rest; drooping eyelids and double vision are often first. Watch chewing and swallowing, and breathing, since chest-wall weakness can impair it. Source: MedlinePlus Medical Encyclopedia, *Myasthenia gravis* (reviewed Apr. 16, 2025).

TRAP
ATP is needed to **release** myosin from actin, not only for the power stroke. No ATP means locked cross-bridges (rigor mortis), not relaxation.

TRAP
During contraction the **sarcomere** shortens; actin and myosin filaments keep their length and slide.

TRAP
Striated does not mean voluntary: cardiac muscle is striated **and** involuntary.

Nervous System and Special Senses

Key structure	Function
CNS	Brain and spinal cord, in the dorsal body cavity, protected by the skull and vertebrae
PNS	12 pairs of cranial nerves and 31 pairs of spinal nerves; every spinal nerve is mixed (sensory and motor)
Neuron	Dendrites carry impulses toward the cell body; the single axon carries them away to the synaptic bulbs
Myelin	Made by Schwann cells (PNS) and oligodendrocytes (CNS); gaps are nodes of Ranvier; myelinated fibers form white matter
Neuroglia	Support, nourish and protect neurons; far more numerous, and unlike neurons they can divide
Meninges	Dura mater (tough, outer), arachnoid (cerebrospinal fluid in the subarachnoid space below it), pia mater (on the surface)
Cerebrum	Two hemispheres joined by the corpus callosum. Frontal: motor cortex, planning, Broca's area (speech production); parietal: touch and body sense; temporal: hearing, long-term memory; occipital: vision
Diencephalon	Thalamus relays sensory input except smell to the cortex; hypothalamus runs homeostasis, the autonomic system and the pituitary
Brainstem	Midbrain, pons, medulla oblongata; the medulla continues as the spinal cord and carries all ascending and descending tracts
Cerebellum	Coordination: compares motor commands with sensory feedback and sends corrections
Spinal cord	From the foramen magnum to the first lumbar vertebra; 31 segments; conduction pathway and reflex center
Autonomic NS	Sympathetic (thoracolumbar): heart rate and breathing up, blood to skeletal muscle, digestion down, more sweat. Parasympathetic (craniosacral, vagus nerve): rest and digest

Physiology essentials

- **Three functions:** sensory input from receptors → integration in the brain → motor output to effectors (muscles and glands).
- **Resting potential:** about −70 mV inside a typical neuron, kept by the Na^+/K^+ pump, which uses ATP.
- **Action potential:** at threshold (about −55 mV) voltage-gated Na^+ channels open and Na^+ rushes in (depolarization); K^+ then flows out (repolarization).
- **Autonomic transmitters:** preganglionic fibers of both divisions release ACh; most sympathetic postganglionic fibers release norepinephrine; parasympathetic postganglionic fibers release ACh.
- **Repair:** peripheral fibers have a neurilemma that helps them regenerate; CNS fibers lack it and do not regenerate.

Special sense	Receptors and pathway
Taste (gustation)	Taste buds detect 5 tastes: sweet, salty, sour, bitter, umami; signals run in the facial, glossopharyngeal and vagus nerves
Smell (olfaction)	Olfactory neurons high in the nasal cavity send axons through the cribriform plate to the olfactory bulb; the only sense that reaches the cortex without a thalamic relay
Hearing (audition)	Eardrum and ossicles pass sound to the cochlea; hair cells of the organ of Corti respond, high pitches near the base, low near the apex; range about 20 to 20,000 Hz
Balance (equilibrium)	Utricle and saccule (otoliths): head position and linear acceleration; semicircular canals: rotation; vestibulocochlear nerve
Vision	Retina: rods for dim light, three kinds of cones for color; the fovea gives the sharpest vision; the optic disc is the blind spot

CLINICAL NOTE Stroke
Blood flow to part of the brain is lost and brain cells start dying within minutes. About 80% of strokes are ischemic (a clot); the rest are hemorrhagic (a vessel bursts). Warning signs come on suddenly: numbness or weakness of the face, arm or leg, especially on one side; confusion or trouble speaking; vision loss; trouble walking or balancing; severe headache with no known cause. Call 911 right away: quick treatment can save a life and improve recovery. Source: MedlinePlus, *Stroke* (updated July 28, 2025).

TRAP
The thalamus relays every sense to the cortex **except smell**.

TRAP
Dendrites carry impulses **toward** the cell body; the axon carries them **away**.

TRAP
7 cervical vertebrae but 8 pairs of cervical nerves: 31 spinal nerve pairs in all (8 C, 12 T, 5 L, 5 S, 1 Co).

Endocrine System

Gland	Hormones and actions
Hypothalamus	Releasing and inhibiting hormones reach the anterior pituitary through the hypophyseal portal veins; its neurons make ADH and oxytocin
Anterior pituitary	GH (growth, protein synthesis), TSH (thyroid), ACTH (adrenal cortex, cortisol), FSH (gamete production), LH (gonadal androgens), prolactin (milk production)
Posterior pituitary	Stores and releases ADH (water reabsorption by kidney tubules) and oxytocin (uterine contraction, milk ejection); controlled by nerve signals
Pineal gland	Melatonin: daily body rhythms, reproductive development
Thyroid	T_4 (about 95% of active thyroid hormone) and T_3, both need iodine; calcitonin from parafollicular cells lowers blood Ca^{2+}
Parathyroids	4 small glands on the back of the thyroid; PTH raises blood Ca^{2+} and is its main regulator
Adrenal cortex	Steroid hormones: aldosterone (keeps Na^+ and water), cortisol (raises blood glucose), small amounts of sex hormones
Adrenal medulla	Epinephrine and norepinephrine, released on sympathetic stimulation during stress
Pancreatic islets	Alpha cells release glucagon when glucose is low; beta cells release insulin when glucose is high

Physiology essentials

- **Ductless glands:** hormones go straight into the blood and reach the whole body, but act only on cells that carry a matching receptor.
- **Two chemical classes:** protein-based hormones (most of them) bind surface receptors and act fast; steroids (adrenal cortex, sex hormones) bind receptors inside the cell and act more slowly, through protein synthesis.
- **Control:** negative feedback for most hormones; hormonal stimuli (a tropic hormone from one gland makes another gland secrete); neural stimuli (e.g., sympathetic nerves and the adrenal medulla).
- **Feedback axis:** TRH (hypothalamus) → TSH (anterior pituitary) → thyroid hormones; high thyroid hormone levels lower TRH and TSH.
- **Iodine:** without enough iodine the thyroid cannot make T_4 and T_3 and may enlarge (goiter).

CLINICAL NOTE Low blood sugar (hypoglycemia)
Blood glucose below 70 mg/dL (3.9 mmol/L). In diabetes it can follow too much insulin in the blood, or exercise in people taking insulin or some other diabetes medicines. Signs: shaking, sweating, fast heartbeat, hunger, irritability, headache, blurred vision, unclear thinking; severe episodes cause seizures or loss of consciousness, and some people feel no warning at all. Treatment uses fast sugar (a sugary drink or food, or glucose tablets) per protocol; an episode that does not improve, or a person who becomes less alert, needs emergency care. Source: MedlinePlus Medical Encyclopedia, *Low blood sugar* (reviewed Jan. 25, 2026).

TRAP
ADH and oxytocin are **made in the hypothalamus**; the posterior pituitary only stores and releases them.

TRAP
Ca^{2+}: PTH **raises** it, calcitonin **lowers** it. Glucose: glucagon (alpha cells) and cortisol **raise** it, insulin (beta cells) **lowers** it.

TRAP
Steroid hormones act **inside** the cell (slower); protein hormones act on surface receptors (faster).

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